

Jonathan Nadeau

Data analyst who builds the fix, not just the report

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Dear Hiring Manager,

I am a data analyst and developer with a mathematics degree from UC Santa Barbara. At Escape Communications, a defense electronics manufacturer in Torrance, I take a problem from a conversation with an engineer to a tested, documented tool that the team uses every day. I have built twelve of them so far, and the one I finished most recently shows how I work better than any list of skills.

Our engineers define every command interface for our embedded software in a JSON file that already generates the C++ code. To verify that code, they were reading raw JSON objects, and the Word documentation was written by hand, which meant it could quietly drift from what the code actually did. I built PAMDocGen, a Python tool that reads that same JSON and produces formatted Word tables they can check in minutes, so the document and the code come from one source and cannot disagree. Our CTO now uses it. The requesting engineer had written a reference document by hand before any code existed, and I wrote 139 tests that compare my output against it element by element. Midway through, a stray save overwrote that reference with a copy of my own tool's output. Every test would have kept passing while proving nothing. A file-integrity check I had built for exactly that case caught it, and I restored the original and verified it before moving on.

Passing the answer key is not the same as being correct. You have to know what the answer key cannot see.

That instinct runs through all my work. When I rebuilt six undocumented quality reports from twenty old workbooks and sixteen database tables, I reproduced four years of results exactly, and in the process found four material errors in the audit record, including one figure overstated 3.4 times that had been carried forward for two years. Nobody asked me to look. The numbers did not reconcile, so I kept going until they did.

I also build for the people on the other end. When I made a reporting tool for 19 years of lab test logs, I realized the data lived on the one computer that runs the tests, so I redirected the test application to the shared drive and now anyone can run reports from their own desk. My tools give plain-language errors instead of stack traces and ship with user guides written for non-developers. I take full ownership of what I am handed, I solve what I can on my own out of respect for my coworkers' time, and I bring leadership the few questions that decide the direction of the work.

I use AI as a development accelerator, deliberately. Much of our data is export-controlled, so I have AI help write Python programs that process it locally and never upload it anywhere. I plan the work, set the scope, catch the errors AI makes, and check every result myself. Along the way I have learned the domain around me: SATCOM terminals, FPGA and embedded software, C/C++, board design, and our ERP inventory data. Next I am completing UCLA Extension's Data Science Certificate, building toward machine learning.

Technology lets people build together what no one could build alone, and I want to be part of where it goes next. I work hard and I work smart, and I know I could make a big impact on your team.

Sincerely,

Jonathan Nadeau